Post-Fire Recovery of Topanga Canyon Lagoon: Monitoring Benthic Macroinvertebrates and Endangered Tidewater Goby Populations

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Background

- Climate change and land management practices have increased wildfire frequency and severity
- The **January 2025 Palisades Fire** burned 23,680 acres, including the Topanga Canyon Watershed
  - Reduced vegetative cover and slope stability
  - Generated hydrophobic soil conditions, accelerating erosion and mobilizing contaminants into waterways
  - Impacted Topanga Canyon Lagoon, critical habitat for **federally endangered northern tidewater gobies** (*Eucyclogobius newberryi)*
- **Benthic macroinvertebrates (BMI)** mediate nutrient cycling, contribute to food web stability, and act as a primary prey base for tidewater gobies
  - Post-fire alterations to substrate composition and water chemistry are expected to impact BMI assemblages
- Endangered gobies were evacuated from Topanga Canyon Lagoon January 17th and returned July 17th, 2025
  - Understanding the rate of prey population recovery is key to making decisions on reintroducing endangered species







Left: Topanga Canyon Lagoon immediately after the Palisades fire
Right: Topanga Canyon Lagoon 20 months after the Palisades fire

Methods

- Substrate samples were collected according to the Surface Water Ambient Monitoring Program (SWAMP) standard operating procedure (SOP)[1] by Resource Conservation District of the Santa Monica Mountains (RCDSMM) staff
- Samples preserved in 70% ethanol processed in a wet laboratory setting by Summer Undergraduate Research Fellows
- BMI specimens were sorted, identified according to taxa, and counted under magnification (4.5x)





Left: The student research team on a SWAMP data collection excursion with RCDSMM
Right: Devan and Chris working on BMI samples in the wet lab

Results

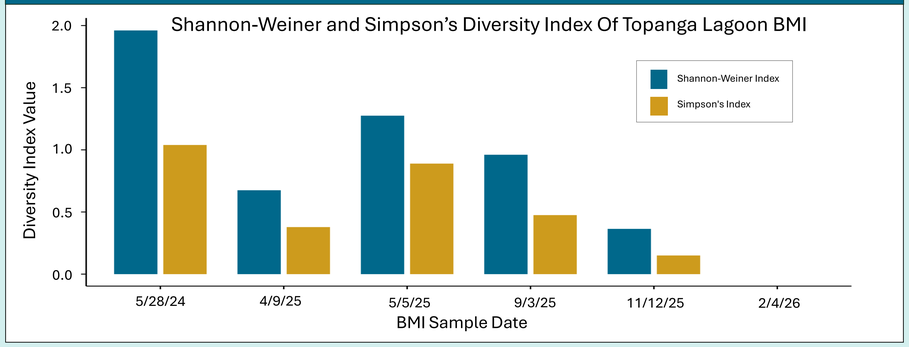


Figure 1: Shows two diversity indices for the BMI samples, which shows the heterogeneity on a scale of 0 - 4.3 for Shannon-Weiner and 0 - 1 for Simpson's Index

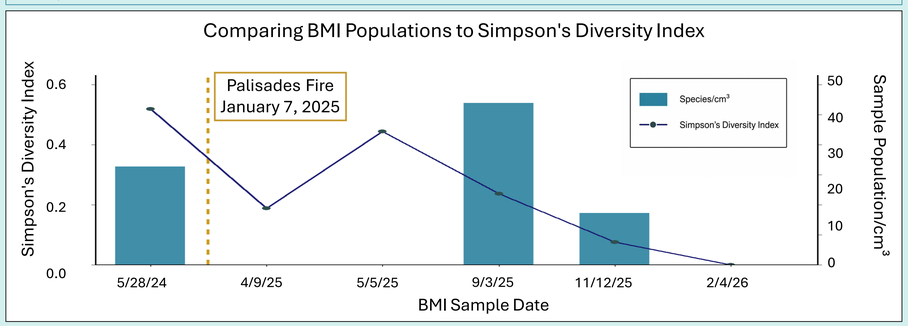


Figure 2: A comparison of the Simpson's Diversity Index to the estimated number of individuals per cm^3 in each sample

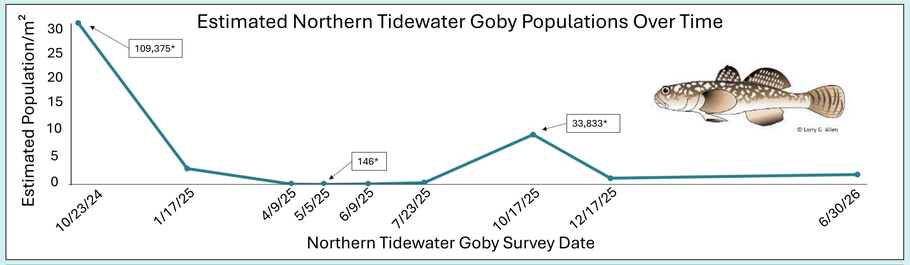


Figure 3: Shows the estimated population of northern tidewater gobies over time. An illustration of a northern tidewater goby is pictured in the upper right of the graph. *(Actual Estimated Population Size)

Representative Benthic Macroinvertebrates

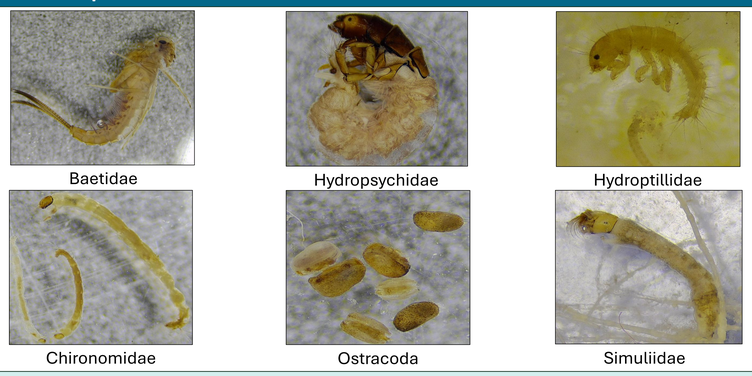
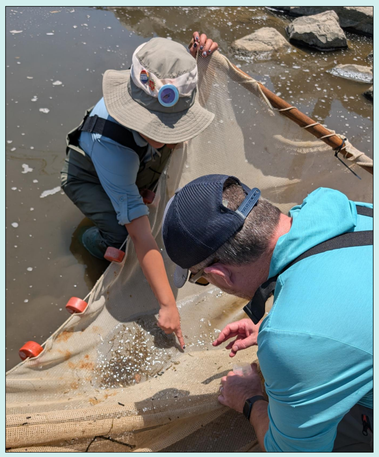


Figure 4: This table pictures some of the most common species identified in Topanga Lagoon BMI samples

Discussion

- BMI species richness declined from **20 taxa pre-fire to as few as 2 taxa immediately after the Palisades Fire** and remained below pre-fire levels throughout the monitoring period
- Species diversity (**Shannon H'**) decreased from **1.74 pre-fire to 0.34 post-fire**, indicating a substantial loss of biodiversity and a simplified macroinvertebrate community
- **Ostracods** were the first species to recolonize after the January 2025 fire, becoming the dominant pioneer taxon during recovery
  - Rapid increase in abundance likely provided an important food source for the endangered northern tidewater goby, maintaining prey availability despite the loss of many pollution-sensitive taxa
- **Hydropsychidae, a pollution-sensitive indicator**[2] of good water quality, were present before the fire but have not returned, suggesting the lagoon has not yet fully recovered
- The **BMI population trend closely mirrors the goby population** with a stark drop post- fire and initial recovery spike beginning in fall 2025







Left: Tatyana and Dr. Spies count gobies in a seine net to estimate the population
Top right: Chris and Dr. Spies pull a seine net through Topanga Lagoon
Bottom right: Adult Northern tidewater goby in a viewing tank

Future Study

- Continue monthly bioassessments to monitor long-term ecological recovery in Topanga Lagoon
- Expand the dataset to improve the understanding of post-fire watershed recovery and long-term ecosystem trends
- Investigate how rainfall events influence post-fire recovery and benthic macroinvertebrate community resilience

Acknowledgements

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Literature Cited

1. P. R. Ode, A. E. Fetscher, and L. B. Busse. 2016.
2. H. Ratia, K. Vuori , A. Oikari. 2012.